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# A Machine Learning-Based Comparative Approach to Predict the Crop Yield Using Supervised Learning With Regression Models

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## Abstract

Agriculture is essential for the existence of all human beings. It is not only a need, but it also contributes to national growth. Agriculture is the only industry that benefits both itself and the rest of the country. It not only offers food and raw materials to a substantial section of the people but also work prospects. Agriculture is the principal source of income for the majority of people in India. It has been in the country for thousands of years, and modern technology and equipment have substituted ancient farming practices. With the advancement of technology in every industry, it has become increasingly vital to incorporate technology into agriculture. Machine learning facilitates the study of large volumes of data and may deliver faster and more precise results, which can aid in the identification of lucrative possibilities and risky threats. The major goal of this research project is to develop a Machine Learning (ML) model to predict farm production. To estimate crop yields, the data was collected and trained using supervised machine learning with six distinct regression models. With a Mean absolute error (MAE) of 468.16 and a Cross-Validation score of 0.6087, Random Forest Regressor outperformed the other models.

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## 1. Introduction

Agriculture provides food, clothes, and shelter. It supports 58% of India's population as it is the major source of income. Due to its great value-adding potential, especially in the food processing business, the Indian food sector is growing rapidly and contributing more to global food commerce every year. Retail accounts for 70% of all sales in the world's sixth-largest economy. The industry accounts for 32% of India's food market and ranks 5th in production, consumption, export, and growth [1]. Since its origin, agriculture has been the primary and preeminent activity of every societal structure throughout human history. It is not only a significant component of the expanding economy but also fundamental for survival. It also contributes a substantial share of employment. As time progresses, the need for manufacturing has multiplicatively expanded [2]. The issue facing the Indian agriculture industry is the incorporation of technology to get the desired results [3].

The new agricultural system must be efficient, robust, varied, and long-term in nature, and it must grow swiftly. With the aid of data acquired, Artificial Intelligence (AI) and Machine Learning (ML) carry the potential in tackling the issues of this new paradigm [4]. These methods help agricultural supply networks make better data-driven decisions [5]. Artificial intelligence is changing the way people generate, deliver, and consume food. In the entire food supply chain, scientists use AI-powered techniques to provide support and insight into sustainable farming, crop protection, infection control, optimal harvesting, food preferences, logistics, food security, and other agricultural-based tasks [4]. Artificial intelligence's flexibility, superiority, and cost-effectiveness are the main reasons for its use in agriculture. These inventions reduce the amount of water, pesticides, and herbicides used; they also aid in the efficient use of labor and increase the quality of the product [6].

This research focuses on leveraging technologies such as Machine Learning (ML) to estimate agricultural yields. Machine learning is a recent technology that is assisting farmers in reducing farming losses by offering detailed crop suggestions and insights [7]. Machine learning is a rapidly expanding strategy that is assisting every industry in making feasible judgments to develop the most prominent of its applications. The fundamental idea is to use Machine Learning models to boost the productivity of the agricultural sector. Most modern gadgets are aided by models that are studied prior to deployment. The fundamental idea is to use Machine Learning models to boost the turnover of the agricultural industry [2]. When there is a loss under adverse circumstances, the ML algorithms may also aid to boost the crop yield production rate. The crop selection approach uses machine learning algorithms to decrease crop yield production losses, regardless of the interfering environment [8].

The yield prediction issue here is a data mining issue, more precisely, a multi-dimensional regression issue [9]. In addition to all of these issues, there is a lot of unpredictability in crop production. The goal of the study is to develop a technique that will not only automate agricultural procedures but also enable farmers to make an educated guess on crop output. Finding relationships between yield (the dependent variable) and other independent factors like temperature and rainfall is the main idea employed in this research work [10]. Regression analysis is significant because it makes it possible to identify the variables that are most important, those that can be overlooked, and the relationships between those variables. It incorporates predicting potential possibilities and liabilities in order to assist farmers in estimating crop production. Machine learning (ML) is a key technology for solving the agricultural production challenge. Supervised Learning predicts ML outcomes from explanatory factors. It gives us an estimate of the best-predicted yield, allowing us to choose the crop that can be grown given the weather [11].

The paper is structured as follows: In the following section, a comprehensive review of literature is performed to understand the research contributions of the existing work based on machine learning (ML) algorithms for agricultural crop yield prediction. The subsequent section is the methodology, which identifies the research problem and describes each phase of the study's proposed approach. Following that comes the results section, which comprises the comparison analysis for assessing regression models and selecting the most appropriate model for final forecasts. The report summarised with the conclusion and suggestions for future scope of the research.

## 2. Literature Review

This study by Anakha Venugopal, Aparna S, et al. [2] uses Machine Learning Classification Models to forecast crop output based on temperature, rainfall, and area. Classifier models include Logistic Regression, Naïve Bayes, and Random Forest. Precision farming should emphasize quality above environmental factors. It lists 14 districts in Kerala's crop names, regions, yields, temperatures, rainfall, humidity, and wind speed. User registers through a mobile app and inputs location and region into the database. Current weather data from OpenWeatherMap

is used to predict yield. Logistic Regression model accuracy was 87.8% and Naive-Bayes was 91.58%. Random Forest Classifier's 92.81% accuracy is the highest among classification methods since it uses Bagging. The accuracy gained after using the Random Forest model to evaluate the test data was 91.34%. This study is done so that crops can be grown more efficiently and effectively.

This research by A. Nigam, S. Garg, et al. [3] evaluates machine learning approaches by incorporating contradictory rainfall and temperature data. Temperature and rainfall are sequential data, therefore the authors considered two-time series models: basic Long-Short Term Memory (LSTM) and Recurrent Neural Network (RNN), ensemble learning models like Random Forest Classifier and XGBoost, and Machine Learning algorithms like K-Nearest Neighbour Classifier (KNN), Logistic Regression, Linear Regression, and Artificial Neural Networks. Temperature and rainfall data span 100 years and are grouped by months, districts, and seasons. Production data from the previous 17 years was cleaned and integrated with current information based on districts, crops, seasons, and regions. LSTM predicted temperature better than Simple RNN. Simple RNN outperformed LSTM in rainfall predictions. Random Forest Classifier identified crop names with 67.80% accuracy, followed by XGBoost, KNN, and Logistic Regression. Random Forest Regressor has the lowest Mean Absolute Error. The Random Forest Regressor best estimates crop production, whereas the Random Forest Classifier, performs best when all parameters are combined. Simple RNN surpasses LSTM for Temperature, but not Rainfall.

This paper by D. J. Reddy and M. R. Kumar [8] compares the benefits, limitations, and prediction metrics of each crop yield prediction model using Machine Learning algorithms. Predicting crop output involves considering soil properties, humidity, nutrients, field data, solar information, wind speed, and pressure. The previous research employed Support Vector Machine (SVM), Convolutional Neural Network (CNN), Artificial Neural Network (ANN), Multiple Linear Regression (MLR), and Deep Neural Network (DNN). According to research, machine learning algorithms are complex and expensive. Second, since Linear Regression only records linear parameter relationships, it can't reliably anticipate non-linear data. Nonlinearity and predictable problems have degraded the KNN yield prediction model. To solve these challenges, reduce the number of input components to make ANN training easier. A huge soil dataset may be used to train an ML system for more accurate crop production forecasting. Each study's crop yield prediction was based on geological position, size, and crop features. Extra features don't necessarily boost accuracy. The investigation assessed the best-performing traits. Combining machine learning with agriculture improved crop yield predictions. Climate change effects on agriculture feature selection could be improved. Fertilizer data should be included in crop yield predictions so farmers can make better decisions in case of low estimates.

The study by Shah A., Dubey A., et al. [10] seeks to develop a strategy for automating agricultural activities and predicting crop production by identifying relationships between yield factors such as temperature and rainfall. Because crop productivity is affected by a variety of environmental factors and may vary under harsh situations, the author ruled out Linear Regression. To anticipate crop yield, they used Multivariate Polynomial Regression, which provided an R-squared value of 0.89 and a Root Mean Square Error (RMSE) value of 9.4. The authors used Random Forest Regression and Support Vector Machine Regression to improve accuracy. With an RMSE of 5.48 and an R-squared value of 0.968, SVM Regression surpassed all trained models and was used in the final approach to predict crop yield.

The research by V. Sellam and E. Poovammal [12] aims to determine if environmental factors like Area Under Cultivation, Annual Rainfall, and the Food Price Index affect crop yield. The dataset covers a period of 10 years, from 1990 to 2000, and includes environmental characteristics such as Area Under Curve (AUC), Annual Rainfall (AR), and Food Price Index (FPI). The analysis used machine learning algorithms, especially Linear Regression. Rice was chosen because it's the most widely grown crop in India. The rice crop used linear regression three times to link AR, AUC, FPI, and Yield. The R2 value for the analysis was 0.72. The linear correlations showed that all three affected crop yield. Regression Analysis from AR to FPI is used to create a connection, which is then combined with the individual relationships to provide a yield. AR affects AUC, which affects crop FPI and production. This study can be expanded by considering additional factors that affect crop yields, such as Minimum Support Price (MSP), weather conditions, soil parameters, etc., and by using data mining and analytical tools to analyze these elements.

This research by P. S. Nishant, P. Sai Venkat, et al. [13] focuses on basic parameters such as state, district, season, and region, which farmers can easily understand. They employed Kernel Ridge, Lasso, Elastic Net, and Stacking Regression to enhance their predictions. It adds a meta-model and uses out-of-fold predictions from prior models to train it. To accomplish so, the team builds a training and validation dataset, trains the chosen basic models

on the training set, and tests them on the validation set. Repeat until desired stacks are obtained. The authors averaged all base model predictions on test data for the predictive phase and used them as meta-features for the meta-model. This study used Lasso Regressor. In this example, Root Mean Square Error (RMSE) was employed. Independently, Elastic Net, Lasso, and Kernel Ridge had a loss of 4%, 2%, and 1%. The loss, however, was reduced to less than 1% after using Stacked Regression. This also demonstrates that Stacked Regression can assist improve accuracy.

This article by Manoj G S1, Prajwal G S2, et al. [14] aims to gather, store, and analyze data, then use machine learning to estimate yield and optimize predictions. This article predicts crops for low, moderate, and high-producing states using multiple regression, decision tree regression, polynomial regression, and K-means clustering. The information includes State, District, Area, production, production area, latitude, longitude, and season. Multiple regression compares independent and dependent variables. This model's R2 accuracy was 28%. Decision Tree Regression generates a tree-shaped regression or classification model by splitting data into smaller groups. Unlike Multiple Regression, the scatter plot for area vs production reveals that the real and estimated values are relatively similar, and Decision tree regression had a 95.7% accuracy score. Due to data, each state's tree was made separately. KMeans places each observation in the nearest mean cluster or centroid. Low-production states include Bihar, Gujarat, Haryana, Madhya Pradesh, Rajasthan, and West Bengal. Tamil Nadu and Uttar Pradesh were high producers in 1999; Karnataka, Maharashtra, and Punjab were average. The Rolling Mean employs series and graphs to examine prospective estimates. It calculates the total output for each state with a window value, which means calculating the sum of productions for three years, i.e. the previous two years plus the current year. After restoring the rolling mean value, the next five years' average scores are predicted using Polynomial Regression.

This paper by Sonal Agarwal and Sandhya Tarar [15] offers a Python-based system to anticipate the most productive crop while decreasing expenditures. This study gives a hybrid method for estimating agricultural yield and the ideal crop for farmers, helping them minimize losses. The hybrid technique employs Support Vector Machine (SVM) and Deep Learning algorithms Long-Short Term Memory (LSTM) and Recurrent Neural Network (RNN). Parameters like pH, relative humidity, and area were also evaluated. The soil's elements, such as nitrogen, phosphorus, and potassium, are obtained from the area of the region. Deep Learning algorithms compute the optimal crop depending on soil, weather, and atmosphere. Wheat, rice, maize, millet, pea, pigeon pea, sugarcane, and green gram are included. Predicting yield uses fertilizers, herbicides, area, UV exposure, and water. The hybrid model (SVM for Machine Learning and LSTM and RNN for Deep Learning) obtained 97% accuracy. This suggests that Deep Learning has a significant role to play in predicting agricultural yields. Based on prediction factors, the algorithm forecasts the most cost-effective crops to farm. AI algorithms assist farmers to reduce soil and weather-related losses using the suggested model.

This study by Miss.Snehal S.Dahikar and Dr.Sandeep V.Rode [16] forecasts agricultural production using Artificial Neural Network (ANN). Adaptation of the ANN is a versatile strategy for selecting important qualities and rejecting costly or problematic ones. Soil type, PH, nitrogen, phosphate, potassium, organic carbon, calcium, magnesium, sulphur, manganese, copper, iron, depth, temperature, rainfall, and humidity were considered for calculating yields. In this research, they examine the feed-forward back propagation neural network because of its ubiquitous applicability. In a feed-forward neural network, neurons are only linked in one direction i.e., forward. Backpropagation uses gradient descent, a supervised training technique, therefore sample inputs and projected outputs must be supplied to the network. The anticipated outputs are compared to the actual outputs for a given input, then the weights of the different layers are modified backward. In ANN, data is trained with random weights and then fine-tuned using backpropagation.

This research by KLS Gogte Institute of Technology et al. [17] compares models to precisely anticipate agricultural output with the least amount of loss. Cotton and Sugarcane yields are estimated using Linear Regression (LR), Decision Tree, and K-Nearest Neighbor (KNN). Yield is determined by soil condition, weather, heavy rain, drought, seed damage, fertilizers, and pesticides. After preprocessing and noise removal, training data is fitted to the model and validated on the test set. Cotton and Sugarcane had the best accuracy with 99% and 98% respectively for the proposed LR model, followed by K-NN and then Decision Tree. The suggested Linear Regression model uses least squares to integrate each point into a polynomial representation and reduce standard errors. Linear regression also propagates external contextual information.

### 3. Methodology

Machine learning is a subfield of Artificial Intelligence (AI) that gives machines the capacity to autonomously learn new things and develop as a consequence of their experiences, even without being specifically programmed to do so [18]. The field of machine learning (ML) has seen an expansion in the number of applications because of advances in computing [19]. Due to its effectiveness in a variety of areas such as forecasting, defect detection, pattern recognition, and so on, machine learning is becoming more frequently employed throughout the world. Yield prediction is a major agricultural challenge. Farmers will be able to predict the yield of their crop using the results of this study before growing it in the cropland, allowing them to make informed decisions. Timely instructions to anticipate future crop output and analysis are essential to help farmers in maximizing agricultural yield. In Fig 1 below, the suggested approach for estimating the crop output of three crops, Bengal Gram, Groundnut, and Maize, for the districts of Telangana, India, is discussed.

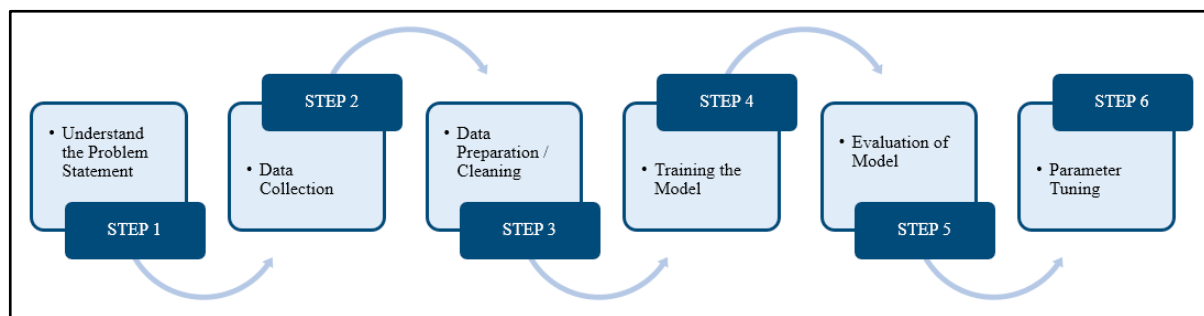


Fig. 1. 6 Step Methodological Framework

#### 3.1 Research Question

The Data was acquired from the Open Data Portal of the State of Telangana, India [20]. The research study is aligned with Zero Hunger as on the Sustainable Development Goals by constructing an optimum model to estimate agricultural output using the most recent information available for the state of Telangana to assist in the decision-making process. In this research work, the authors addressed the following research question:

- How can machine learning techniques be employed to develop an accurate crop yield prediction model for Bengal gram, groundnut, and maize in the Telangana area of India from 2016 to 2018?

#### 3.2 Data collection

Agriculture is a multi-stage process that primarily includes three phases of production. Farmers use water, seeds, fertilizers, pesticides, and insecticides at specific intervals, depending on the natural state of weather conditions. The farm's output is yield, which is impacted by a multitude of elements. A Machine Learning (ML) model can be used to anticipate historical production using these elements, demonstrating statistical relationships between variables that affect crop yield. The study focuses on the agricultural crops maize, groundnut, and Bengal gram, which are significant in Telangana's agricultural ecosystem. In light of this, certain datasets from the Telangana Open Data Portal [20] by the ITE&C Department, Government of Telangana were analyzed. It may provide information about Telangana's agricultural output for three years, notably 2016, 2017, and 2018. For each of the three years, the files contained data on the lowest and highest temperatures for each month in Telangana's 31 districts, as well as annual rainfall and crop yields for the three crops stated above.

#### 3.3 Data preparation

The final dataset that was used to train the model was not extracted in its entirety. It was obtained by individually obtaining information on each of the features from the numerous resources provided by the organizers of the hackathon. Each dataset underwent extensive pre-processing, which included performing crucial tasks like resolving missing values and encoding categorical data. Ultimately, the dataset was divided into training and test sets after all

of the feature variables were incorporated into a single file.

By setting the parameters for numerical and categorical variables individually, the procedures for resolving missing values and encoding were done using a common pipeline. The missing values in numerical data were replaced by the median while missing values in categorical data were replaced by the most frequently occurring value using an Imputer. The OneHot encoder was used to encode the categorical data. It's critical in machine learning to distinguish between independent and dependent variables in a dataset. Crop, District, Year, Total Rainfall, and Maximum and Minimum Temperatures for all months of 2016, 2017, and 2018 are the independent variables in the dataset whereas Total yield is the dependent variable.

One of the most important phases in data preparation is splitting the dataset into Train and Test sets since this will improve the performance of the machine learning model. If the model is appropriately trained and has a high training accuracy, but it is given a new dataset, its performance will decline. As a result, it is important to create a machine learning model that works well with both the training and test datasets. These datasets can be defined as follows: 80% Training Data and 20% Testing Data.

### 3.4 Training the model

After completing the necessary preprocessing and splitting the data into train and test sets, the model was trained on the training dataset. The most crucial phase in machine learning is training. The prepared data is fed to the machine learning model during training so it can detect trends and make predictions. As a consequence, the model learns from the data and can complete the goal assigned. The model improves in predicting over time as it is trained. While predicting a continuous dependent variable from a set of independent factors by applying regression analysis. Six alternative regression models were used to train the data, including Linear Regression, Decision Tree Regression, Gradient Boosting Regression, Random Forest Regression, Xgboost Regression, And Voting Regression.

- *Linear regression:* A linear method to model the connection between a numerical output and one or more explanatory factors (also known as dependent and independent variables) is known as linear regression in statistics. When there are one or more inputs, one may perform an iterative method to enhance the coefficient values by reducing the model's error on the training data. Gradient Descent is the name of the process, and it starts with random values for each coefficient. The coefficients are updated in the direction of reducing the error using a learning rate as a scaling factor. The process is continued until the sum squared error is reduced to a minimum or no more improvement is achievable.
- *Decision tree regression:* A decision tree is a representation of decisions and all of their possible consequences, including outcomes, input costs, and utility, that employs a flowchart-like tree structure. The supervised learning algorithms include the decision-tree algorithm. It may be used with both continuous and categorical output variables. A regression tree is a decision tree that may be used to predict continuous-valued outputs rather than discrete outputs when doing regression. Regression Trees are used for regression and can be beneficial in a variety of situations when the connection between the variables is non-linear. Overfitting is a risk with this method.
- *Gradient boosting regression:* A prominent boosting technique is gradient boosting. Each predictor in gradient boosting corrects the mistake of its predecessor. Gradient boosting is one of the boosting strategies that is used to reduce the model's bias error. The gradient boosting approach may be used to forecast not just continuous (as a Classifier) but also categorical target variables (as a Regressor). A gradient-boosted trees model is constructed in the same stage-wise manner as other boosting approaches, but it differs in that it allows optimization of any differentiable loss function.
- *Random forest regression:* Random Forest is a supervised machine learning algorithm that uses numerous decision trees and a technique called Bootstrap and Aggregation, sometimes known as bagging, to solve classification and regression problems. Every decision tree has a large variance, but when all of them are mixed in parallel, the final variance is minimal since each decision tree is perfectly trained on that particular sample data, and so the outcome is dependent on numerous decision trees rather than one. The ultimate result in a regression problem is the average of all the outputs.
- *Xgboost regression:* XGBoost is a form of ensemble learning. Ensemble learning is an approach for combining the predictive abilities of numerous learners systematically. The result is a single model that combines the output of numerous models. XGBoost anticipates having base learners who are consistently awful at the rest so that when all of the predictions are added together, the bad ones cancel out and the better ones add up to make final

positive predictions.

- *Voting regression*: A voting regressor is recursive that fits many baseline regressors to the entire dataset. The individual predictions are then averaged to give a final prediction.

### 3.5 Evaluation of model

Following training the model, the authors have examined its performance. This is accomplished by putting the model to the test on previously ‘unknown’ data. The unseen data is the testing set that is collected after splitting the data. If the test is on the same data that was used for training, Ideally won't receive an accurate result since the model is already familiar with the data and identifies the same patterns it did before. This will provide a disproportionately high level of precision. Applying the model to testing data will give a better idea of how it will perform and how efficient it will be. For this, three metrics are used to evaluate the model: Mean Absolute Error, R2 Score, and Cross-Validation Score.

- *Mean absolute error*: The mean absolute error (MAE) is computed by adding the absolute differences between the actual and estimated values of each observation over the complete array and dividing the result by the number of observations in the array. The MAE will be used to determine how a collection of estimates' errors vary from one prediction to the next. Lower numbers are preferable because the score is negatively orientated.
- *Mean squared error*: The average squared error, or the difference between the estimated values and real values, is measured by an estimator's Mean Squared Error (MSE). MSE is undefined. Simply expressed, a value of 0 is ideal. Since there's no right answer, the MSE helps choose amongst prediction models.
- *R2 score*: R squared, also known as the coefficient of determination, operates by calculating the amount of variation in the predictions. It calculates the link between the movements of a dependent variable and the movements of an independent variable. It will not tell us if the selected model is excellent or terrible, or whether the data and forecasts are skewed. Furthermore, there is no right answer for what the R2 score should be. 100% denotes a flawless association. There are, however, models with low R2 that are equally useful.
- *Cross-validation score*: Cross-validation is most commonly used in applied machine learning to evaluate a machine learning model's competence on unknown data to prevent overfitting. It is a method of assessing machine learning models that involves training numerous models on subsets of the available input data and then evaluating them on the complementary subset.

### 3.6 Parameter tuning

After constructing and assessing the model, possible ways to increase the accuracy by parameter tuning were explored. This is accomplished by fine-tuning the parameters in the model. Parameters are the variables in the model that are set by the programmer. The accuracy will be the highest for a specific value of the parameter. Finding these settings is referred to as parameter tuning. GridSearchCV was employed on the two models that were previously chosen to find the optimal set of parameters. GridSearchCV uses the Cross-Validation technique to test the model for all possible combinations of the values supplied in the dictionary of parameters. As a result, after applying this function, the accuracy/loss for each combination of hyperparameters was retrieved and the set of parameters that performed best was chosen. After having chosen the final machine learning model, it was used to make predictions on new data instances.

## 4. Results

This study's primary objective is to develop the most effective Machine Learning (ML) model possible for estimating agricultural yields for three different types of crops: Bengal gram, Groundnut, and Maize. It may be beneficial for the farmers to take the required measures to boost crop production, which may also be useful in the process of decision-making. To make accurate predictions of the crop yields, the data was trained on six different kinds of alternative regression models, including Linear Regression, Decision Tree Regression, Gradient Boosting Regression, Random Forest Regression, Xgboost Regression, And Voting Regression. Table 1, displays the values for all 3 metrics: Mean Absolute Error (MAE), R2 score, and Cross-Validation score for all the 6 models mentioned above. These criteria are critical in evaluating models and selecting the best suitable model for the final predictions.

Table 1. Metrics of models trained

| Regression Models            | Mean Absolute Error (MAE) | Mean Squared Error (MSE) | R2 Score | Cross-Validation Score |
|------------------------------|---------------------------|--------------------------|----------|------------------------|
| Linear Regression            | 1121.60                   | 1384.58                  | 0.3470   | 0.2022                 |
| Decision Tree Regression     | 745.75                    | 1015.52                  | 0.4516   | 0.3425                 |
| Gradient Boosting Regression | 532.34                    | 848.88                   | 0.7544   | 0.6009                 |
| Random Forest Regression     | 475.17                    | 851.30                   | 0.7732   | 0.6158                 |
| XGBoost Regression           | 488.32                    | 769.49                   | 0.8334   | 0.5026                 |
| Voting Regression            | 549.02                    | 828.35                   | 0.7599   | 0.5899                 |

#### 4.1 Comparative Analysis

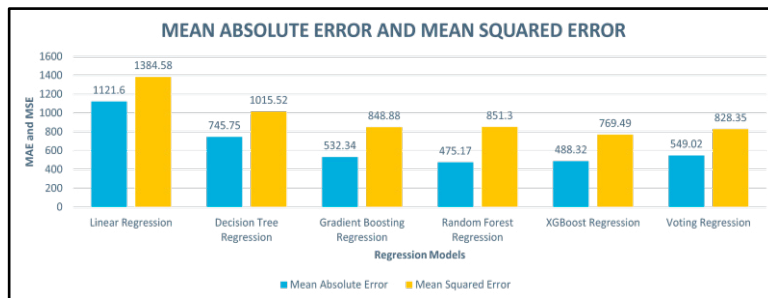


Fig. 2. Graphical representation of Mean Absolute Error and Mean Squared Error for the trained models

The Fig. 2 shows the Mean Absolute Error (MAE) and Mean Squared Error (MSE) for trained models. A 0 value for both MAE and MSE is optimal. The blue graph represents the MAE value and is lowest for Random Forest Regression followed by XGBoost Regression and Gradient Boosting Regression. The yellow graph represents the MSE value which is lowest for XGBoost Regression, followed by Voting Regression, Gradient Boosting Regression, and Random Forest Regression.

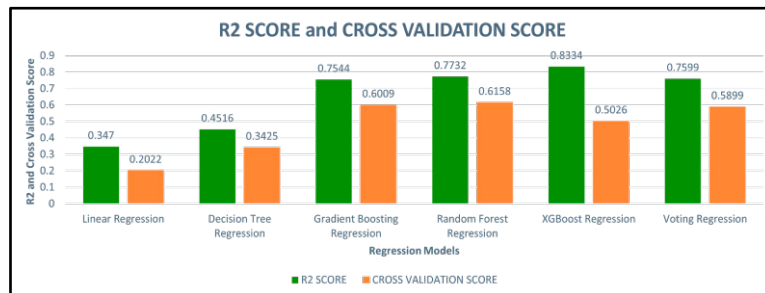


Fig. 3. Graphical representation of R2 Score and Cross-Validation Score for the trained models

Fig. 3. is a graphical representation of the R2 score and Cross Validation score for the trained Regression models. An R2 score of 1 is considered ideal. The R2 score is shown by the green graph, while the cross-validation score is represented by the orange graph. The best cross validation score is for Random Forest Regression, followed by Gradient Boosting Regression, Voting Regression, and XGBoost Regression. The greatest R2 scores are for XGBoost Regression and Random Forest Regression.

It is critical to evaluate and select a final model that is appropriate for the work at hand. Apart from that, it is important to determine if the model is best suited for numerical or categorical data and make the appropriate choice. The Random Forest Regressor and XGBoost Regressor perform the best out of the six models trained with the highest R2 score and lowest MAE value, as seen in Figures 2, 3, and Table 1, therefore these two models are further examined in the following stage.

#### 4.2 Comparison of Models Before and After Hyperparameter Tuning

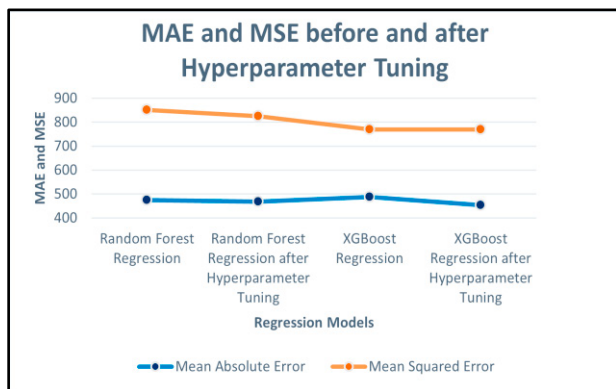


Fig. 4(a). MAE and MSE value comparison, before and after Hyperparameter Tuning

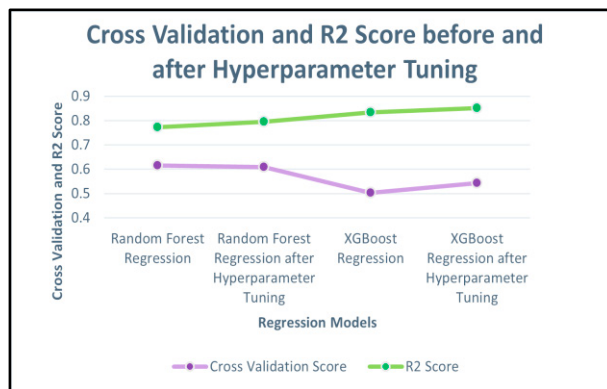


Fig. 4(b). R2 Score and Cross Validation Score comparison, before and after Hyperparameter Tuning

The MAE and MSE values for the Random Forest and XGBoost regressions before and after hyperparameter optimization are compared in Fig. 4(a). After hyperparameter tuning, the MAE and MSE values for Random Forest Regression marginally dropped. The MAE value of the XGBoost Regression slightly decreased but the MSE value was nearly equal after training the models with an optimal set of parameters.

The R2 and Cross Validation scores are compared in Fig. 4(b). before and after hyperparameter optimization. After hyperparameter tuning, the cross-validation score for Random Forest Regression remained essentially unchanged, however it marginally improved for XGBoost Regression. After hyperparameter tuning, the R2 scores for both models improved, with XGBoost Regression having a better value.

Table 2. Evaluation Metrics with the optimal set of hyper-parameters

| Regression Models with Best Hyper-Parameters | Mean Absolute Error (MAE) | Mean Squared Error (MSE) | R2 Score | Cross Validation Score |
|----------------------------------------------|---------------------------|--------------------------|----------|------------------------|
| Random Forest Regression                     | 468.16                    | 825.29                   | 0.7952   | 0.6087                 |
| XGBoost Regression                           | 454.03                    | 770.15                   | 0.8509   | 0.5431                 |

Table 2 shows that XGBoost Regression has a lower MAE and MSE value than Random Forest Regression, although the difference is not significant. When comparing the R2 and cross-validation scores of both models, a greater R2 value and low cross-validation score show that the XGBoost model's predictions are extremely variable, indicating that the model may be overfitting. The cross validation score of Random Forest Regression is greater, indicating that it works better on unknown data. Based on the dataset, Random Forest Regression appears to be the most efficient final model for the research.

## 5. Conclusion and Future Scope

The data for this research was collected from the Open Data Portal of the State of Telangana, India which was listed by the Department of Agriculture and Cooperation, Government of Telangana. The aim of this research study is to analyze how machine learning techniques can be employed to develop an accurate crop yield prediction model for Bengal gram, groundnut, and maize in the Telangana area of India from 2016 to 2018. It is aligned with Zero Hunger as on the Sustainable Development Goals by constructing an optimum model to estimate agricultural yield output using the most recent information available for the state of Telangana to assist in the data-driven decision-making process.

This research heavily emphasized the adaption of the Machine Learning (ML) techniques and trained on six different supervised-based learning regression models to estimate the crop yield, namely Linear Regression, Decision

Tree Regression, Gradient Boosting Regression, Random Forest Regression, Xgboost Regression, and Voting Regression, two of which were the most accurate: Random Forest Regression and XGBoost Regression. Based on our observation, the XGBoost Regression model was overfitting with a high R2 score and lower Cross Validation score. Furthermore, with an MAE score of 468.16, MSE score of 825.29, R2 score of 0.7952 and a Cross-Validation score of 0.6087, the Random Forest Regression model outperformed the XGBoost Regression model after parameter tuning.

Many Artificial Intelligence (AI), Deep Learning (DL) and Computer Vision (CV) techniques can be used on images of the field and crop to determine if they are infected with any illness or the presence of weed, which affects the quality of the crop, and can be separated from the healthy crops as soon as feasible.

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